

# Module 10: Epigenetics — Keys to Success

## Learning Objectives

By the end of this module, you should be able to:

1. Explain why cells with the same DNA look and act differently
2. Define epigenetics and explain how it is different from a mutation
3. Describe how DNA methylation and histone acetylation work
4. Give examples of how diet, stress, or other environmental factors can change gene activity

## The Big Picture

In Module 07 you learned: **DNA → RNA → Protein**. But here's the key question — does every cell make every protein? **No**. A muscle cell only uses muscle genes. A nerve cell only uses nerve genes. Something is deciding which genes to turn on. That "something" is **gene regulation**, and one of the most important forms of it is **epigenetics**.

## Key Terms

### Why Don't All Genes Turn On?

- **Gene expression** — When a cell actually uses a gene to make its protein (remember: transcription + translation from Module 07)
- **Gene regulation** — The process of turning genes on or off
- **Transcription factor** — A protein that attaches near a gene and helps turn it on or off (remember: transcription starts at the **promoter** — Module 07)

**Simple analogy:** Your DNA is a giant recipe book. Gene regulation is deciding which recipes to cook today. Every cell has the whole book, but a liver cell only cooks liver recipes.

## Epigenetics — Changing the Volume, Not the Song

- **Epigenetics** — Changes that turn genes on or off **without changing the DNA sequence**
- This is NOT a mutation — a mutation changes the letters of DNA (Module 07);  
epigenetics changes whether the gene gets *read* or *ignored*
- Epigenetic changes can be **reversed** (mutations usually cannot)

### Two Main Epigenetic Tools

| Tool                       | What It Does                                                               | Memory Trick                             |
|----------------------------|----------------------------------------------------------------------------|------------------------------------------|
| <b>DNA Methylation</b>     | Adds a chemical tag to DNA → gene is <b>silenced</b> (OFF)                 | <b>M</b> ethylation = <b>M</b> uting     |
| <b>Histone Acetylation</b> | Adds a chemical tag to histones → DNA loosens → gene is <b>active</b> (ON) | <b>A</b> cetylation = <b>A</b> ctivation |

- **Histones** are the spool-like proteins that DNA wraps around (like thread on a spool)
- When histones are **tightly wound** → DNA is closed → gene is OFF
- When histones are **loosened** → DNA is open → gene is ON

### X-Inactivation — You Can See Epigenetics

- Females have two X chromosomes (XX). Early in development, one X in each cell is randomly shut down → it becomes a dense blob called a **Barr body**
- **Calico cats** — The patches of orange and black fur happen because different cells randomly shut down different X chromosomes
- This is a real, visible example of epigenetics — same DNA, different outcome depending on which X is silenced

### How the Environment Affects Your Genes

Remember from Module 09 that phenotype = genotype + environment? Epigenetics is one of the main ways environment influences phenotype:

- **Diet** — Nutrients like folic acid provide methyl groups that can silence or activate genes

- **Stress** — Chronic stress can change which genes are active, especially in the brain
- **Exercise** — Can turn on genes involved in metabolism
- **Toxins** — Pollutants can disrupt normal epigenetic patterns

**Your genes are not your destiny.** Even identical twins (same DNA from Module 08 — mitosis produces identical copies) can develop different traits as they age because of epigenetic differences.

## Study Tips

1. **Start from Module 07** — If you understand DNA → RNA → Protein, then epigenetics is just: "what decides whether that arrow gets used or not"
2. **Methylation = Muting** — both start with M; methylation silences genes
3. **Acetylation = Activation** — both start with A; acetylation opens DNA up
4. **Epigenetics ≠ mutation** — Mutations change the recipe. Epigenetics closes the recipe book to that page.
5. **Calico cats** — The single best example to remember; visible proof that same DNA → different outcomes